

Bounded Γ_n forces bounded Δ_n : a negative answer to Remark 5.5 of arXiv:math/0610427

Candidate full disproof of the conjectured existence statement

17 August 2026

Status: candidate full disproof, high confidence. For every probability measure P considered in the source paper,

$$\|\Delta_n(P)\|_\infty = \|\Gamma_n(P)\|_{2 \rightarrow \infty}^2 \leq \|\Gamma_n(P)\|_2^2.$$

Hence a uniformly bounded sequence $\|\Gamma_n(P_n)\|_2$ can never coexist with an unbounded sequence $\|\Delta_n(P_n)\|_\infty$. The sequence conjectured in Remark 5.5 does not exist.

1 The source question

Kontorovich associates to a probability measure P on a product space an upper-triangular matrix $\Delta_n(P)$ whose diagonal entries are one and whose strictly upper-triangular entries are mixing coefficients $\bar{\eta}_{ij} \in [0, 1]$. Thus

$$\|\Delta_n(P)\|_\infty = \max_{1 \leq i < n} \left(1 + \sum_{j=i+1}^n \bar{\eta}_{ij} \right). \quad (1)$$

The paper also uses Samson's matrix $\Gamma_n(P)$, defined by taking the entrywise nonnegative square root,

$$(\Gamma_n(P))_{ij} = \sqrt{(\Delta_n(P))_{ij}}, \quad (2)$$

and writes $\|\Gamma_n(P)\|_2$ for its induced Euclidean (spectral) operator norm.

Remark 5.5 asks whether there is a sequence (P_n) such that $\|\Gamma_n(P_n)\|_2$ is bounded while $\|\Delta_n(P_n)\|_\infty$ is unbounded, and conjectures that such a sequence exists [1, p. 11, Remark 5.5]. The latest arXiv version is v2, dated 14 October 2006, and retains the question.

Remark 5.5. Note that for the first direction in the proof of Theorem 5.3 we constructed a sequence of measures $\mathbf{P}_n$ such that $\|\Delta_n(\mathbf{P}_n)\|_\infty = 2$ is bounded while $\|\Gamma_n(\mathbf{P}_n)\|_2 = \Omega(n^{1/2})$. Is there a sequence of measures for which $\|\Gamma_n(\mathbf{P}_n)\|_2$ is bounded and $\|\Delta_n(\mathbf{P}_n)\|_\infty$ unbounded? We conjecture that such a sequence of measures indeed exists, but leave its construction for future investigation.

2 The matrix identity

The obstruction does not require the probabilistic structure of the entries.

Lemma 1. *Let $D = (d_{ij})$ be an $m \times n$ matrix with nonnegative entries, and set $G = (\sqrt{d_{ij}})$ entrywise. Then*

$$\|D\|_\infty = \|G\|_{2 \rightarrow \infty}^2 \leq \|G\|_{2 \rightarrow 2}^2. \quad (3)$$

Here $\|D\|_\infty$ is the induced ℓ_∞ operator norm.

Proof. For any real matrix A ,

$$\|A\|_{2 \rightarrow \infty} = \max_{1 \leq i \leq m} \|\text{row}_i(A)\|_2. \quad (4)$$

Indeed, Cauchy–Schwarz gives the upper bound in (4); equality is obtained by taking the unit vector parallel to a row of maximum Euclidean norm. Applying this to G gives

$$\|G\|_{2 \rightarrow \infty}^2 = \max_i \sum_j |g_{ij}|^2 = \max_i \sum_j d_{ij} = \|D\|_\infty,$$

where the last equality uses $d_{ij} \geq 0$. Finally, $\|y\|_\infty \leq \|y\|_2$ for every $y \in \mathbb{R}^m$, so

$$\|G\|_{2 \rightarrow \infty} = \sup_{\|x\|_2=1} \|Gx\|_\infty \leq \sup_{\|x\|_2=1} \|Gx\|_2 = \|G\|_{2 \rightarrow 2}.$$

Squaring proves (3). □

3 Full negative answer

Theorem 2. *For every n and every probability measure P for which the source matrices are defined,*

$$\|\Delta_n(P)\|_\infty = \|\Gamma_n(P)\|_{2 \rightarrow \infty}^2 \leq \|\Gamma_n(P)\|_2^2. \quad (5)$$

Consequently, if $\sup_n \|\Gamma_n(P_n)\|_2 \leq C < \infty$, then $\sup_n \|\Delta_n(P_n)\|_\infty \leq C^2$. In particular, the sequence requested and conjectured in Remark 5.5 does not exist.

Proof. Apply Lemma 1 with $D = \Delta_n(P)$ and $G = \Gamma_n(P)$, using the entrywise relation (2). Formula (1) omits the last row from its displayed maximum. That row has sum one, while every earlier row also contains its diagonal one, so including the last row does not change the maximum. Thus Lemma 1 applies without any convention mismatch. The uniform implication follows directly from (5). □

4 Checks, sharpness, and relation to Theorem 5.3

Sharpness. For a product measure, $\Delta_n = \Gamma_n = I_n$. Equality holds throughout (5); the constant one cannot be improved.

No hidden measure-theoretic assumption. Essential suprema enter only in defining the numbers $\bar{\eta}_{ij}$. Once the finite matrices have been formed, the proof is an exact algebraic identity. It applies equally to finite, countable, or continuous underlying state spaces whenever the source definitions apply.

Compatibility with Theorem 5.3. The source proves that the ratio

$$R_n = \frac{\|\Gamma_n(P_n)\|_2}{\|\Delta_n(P_n)\|_\infty}$$

can tend either to zero or to infinity. This does not conflict with (5). The zero-ratio construction has both norms unbounded, with Γ_n permitted to grow on the scale of the square root of Δ_n ; the infinite-ratio construction keeps Δ_n bounded while Γ_n grows. The missing fourth possibility—bounded Γ_n and unbounded Δ_n —is exactly what (5) forbids.

5 Literature and version audit

Exact-pharse searches for the question, together with searches combining the author, the two matrix symbols, mixing coefficients, and operator norms, found no explicit published resolution as of 17 August 2026. The author’s 2007 thesis recalls the Γ_n – Δ_n comparison and Theorem 5.3 in Section 5.7, but omits this conjecture and does not state (5) as an answer [2, §5.7]. The related later paper on prescribed η -mixing coefficients addresses a different realization problem [3]. This packet therefore records Theorem 2 as an apparently new explicit negative answer, subject to independent literature review.

References

- [1] L. Kontorovich, *Metric and Mixing Sufficient Conditions for Concentration of Measure*, arXiv:math/0610427v2 (2006), <https://arxiv.org/abs/math/0610427>.
- [2] L. Kontorovich, *Measure Concentration of Strongly Mixing Processes with Applications*, Ph.D. thesis, Carnegie Mellon University (2007), <https://www.cs.bgu.ac.il/~karyeh/thesis.pdf>.
- [3] L. Kontorovich, *Constructing processes with prescribed mixing coefficients*, arXiv:0711.0986 (2007), <https://arxiv.org/abs/0711.0986>.